AI-Based Presentation Skills Evaluation System

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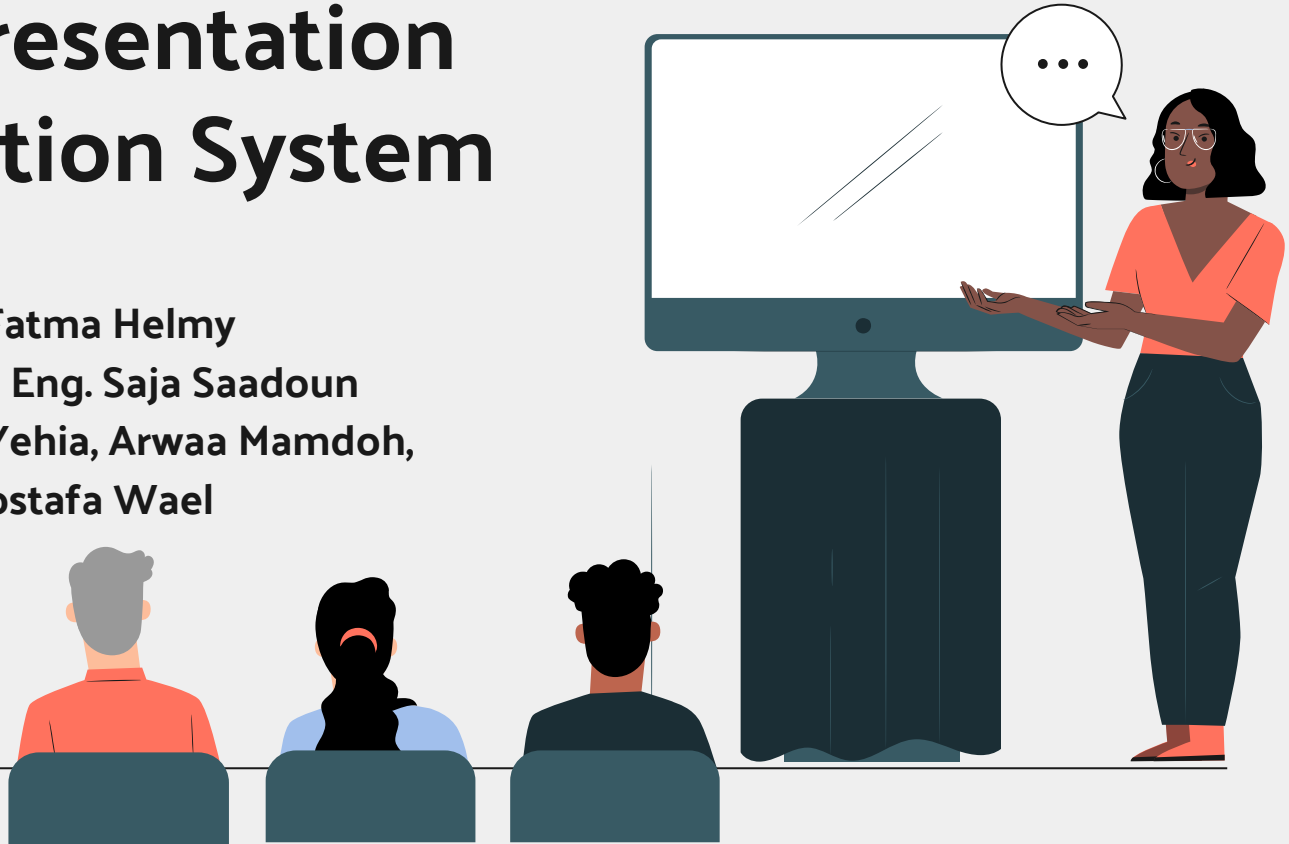

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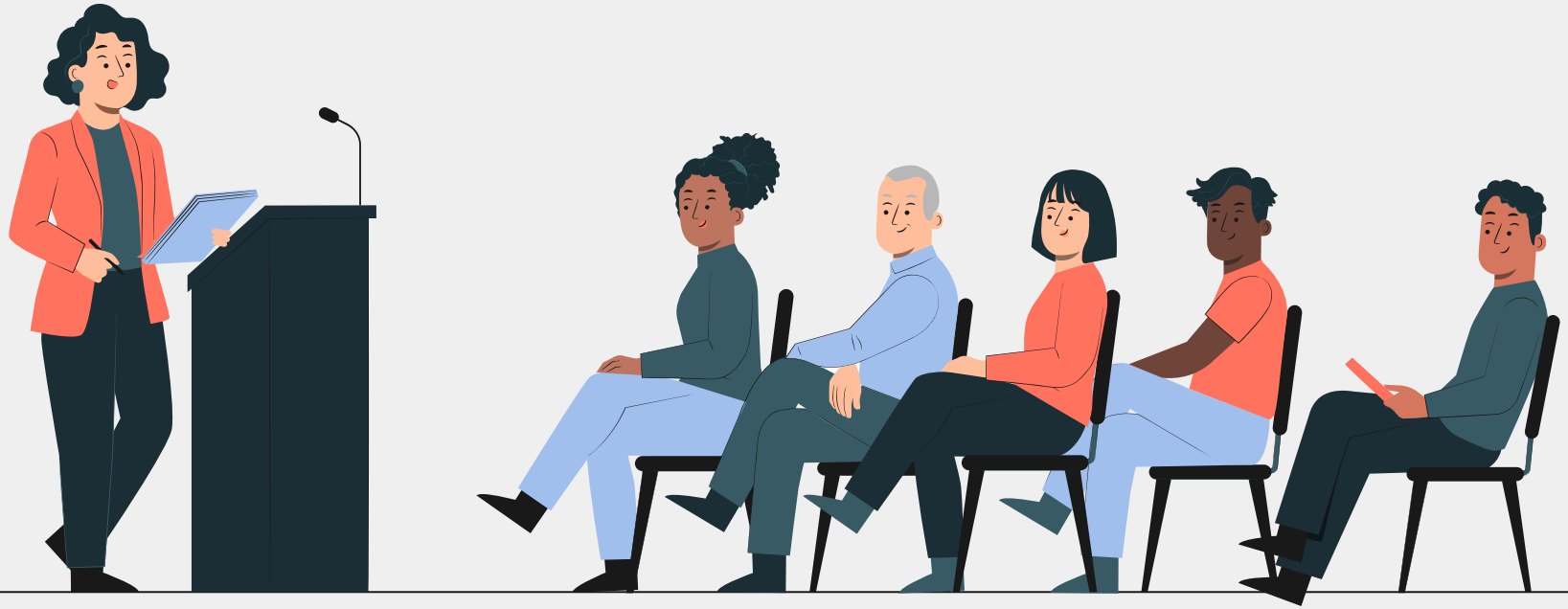
01 INTRODUCTION



01 INTRODUCTION

Problem Statement

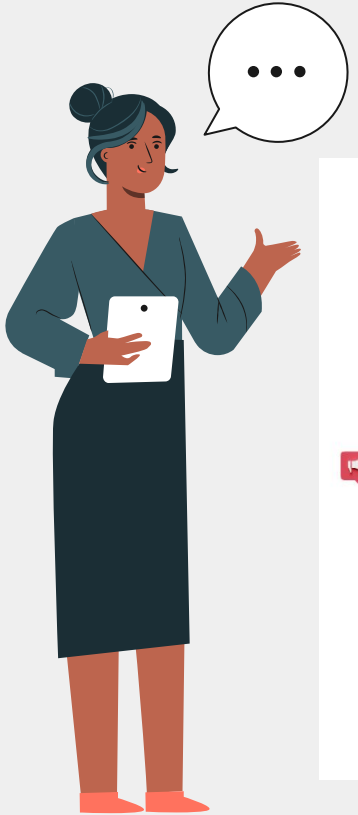
- Many individuals struggle with public speaking due to unclear speech, lack of confidence, and poor body language.
- Traditional training lacks objective feedback and is not scalable. Our system automates the evaluation of presentation skills, offering real-time AI-driven insights to improve performance.

01 INTRODUCTION

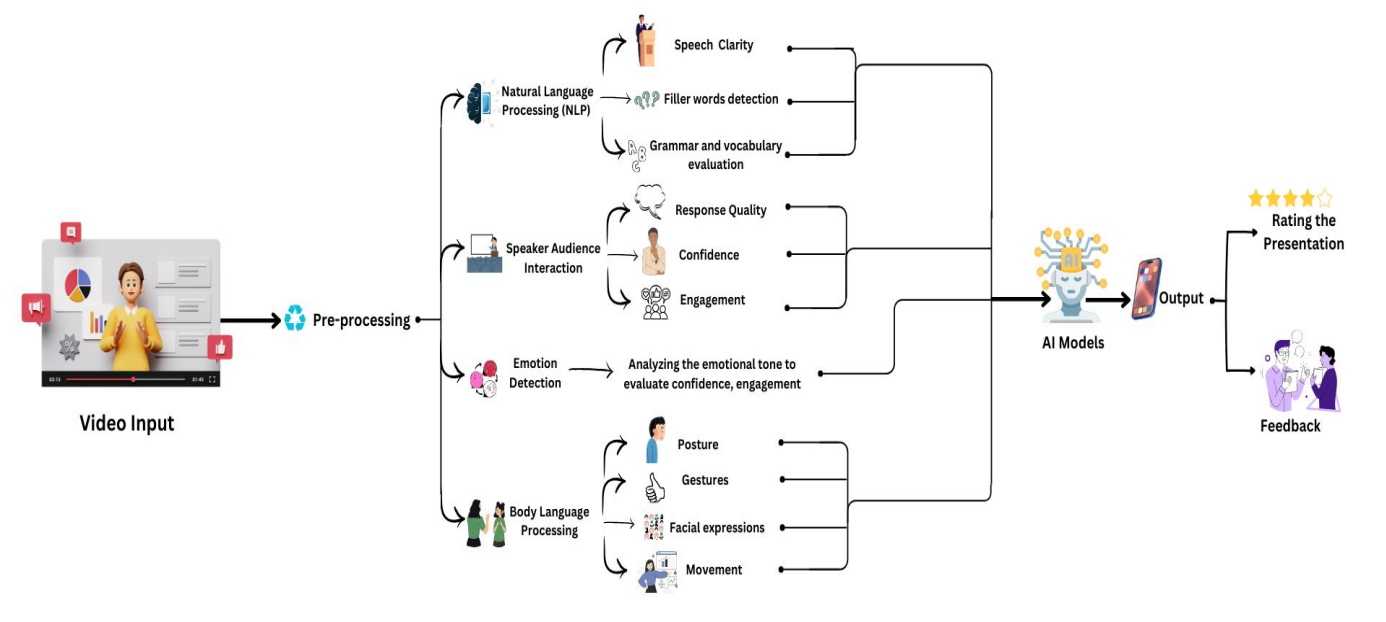
Objectives

- Analyze speech clarity, pace, and grammar.
- Detect body language, gestures, and facial expressions.
- Provide personalized feedback to improve public speaking skills.
- Track user progress over time.

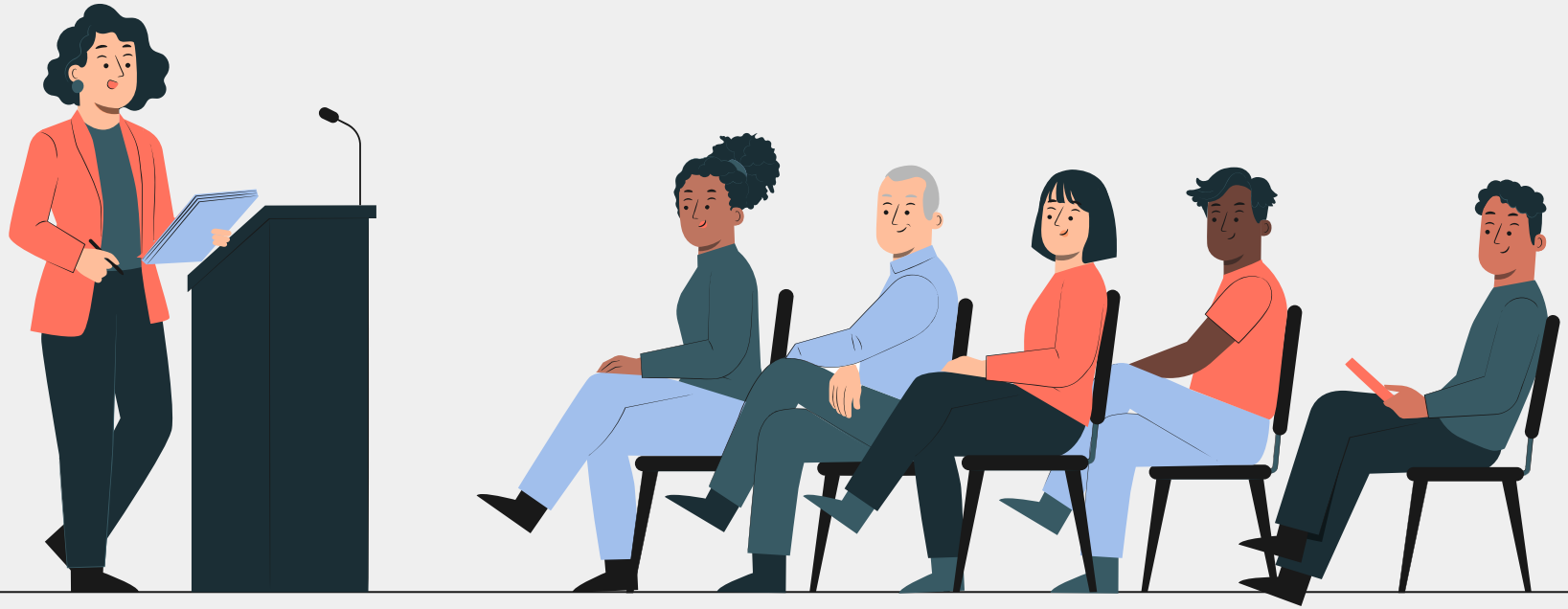
01 INTRODUCTION



System Overview



02 REQUIREMENTS

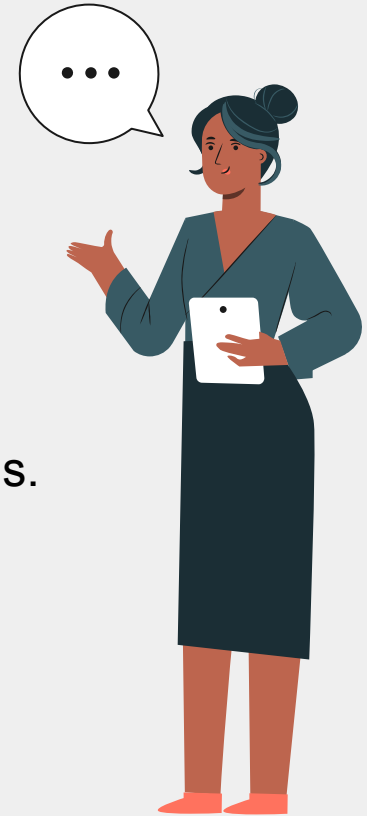


02 REQUIREMENTS

Functional Requirements

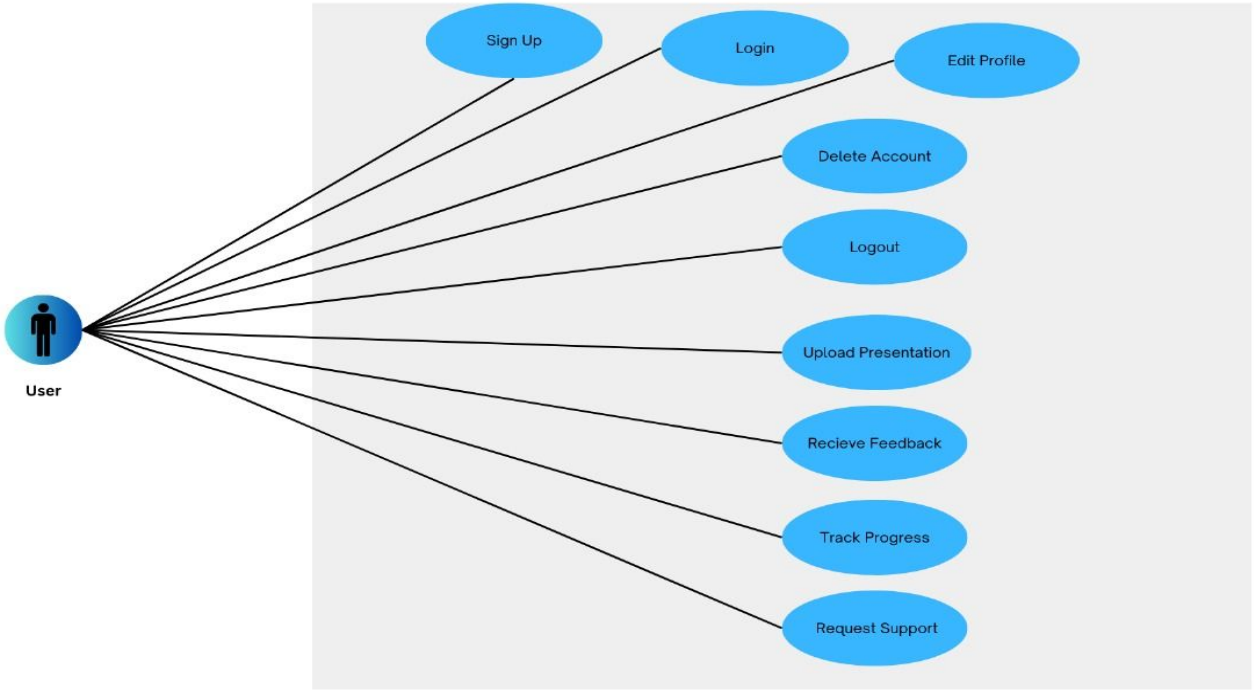
The system must:

- Allow users to upload presentation videos.
- Process speech, body language, and facial expressions.
- Generate feedback reports with improvement suggestions.
- Support multiple users and history tracking.



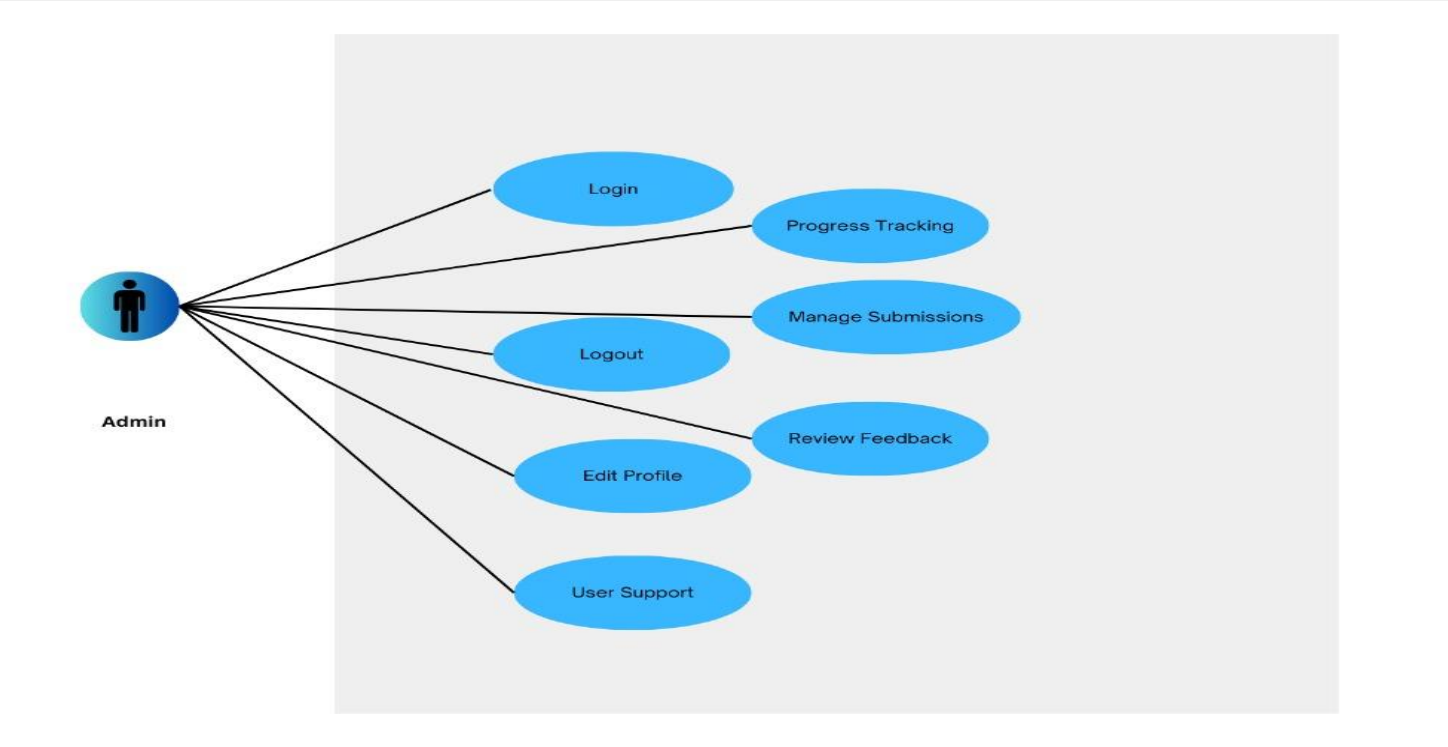
02 REQUIREMENTS

User Use Case Diagram



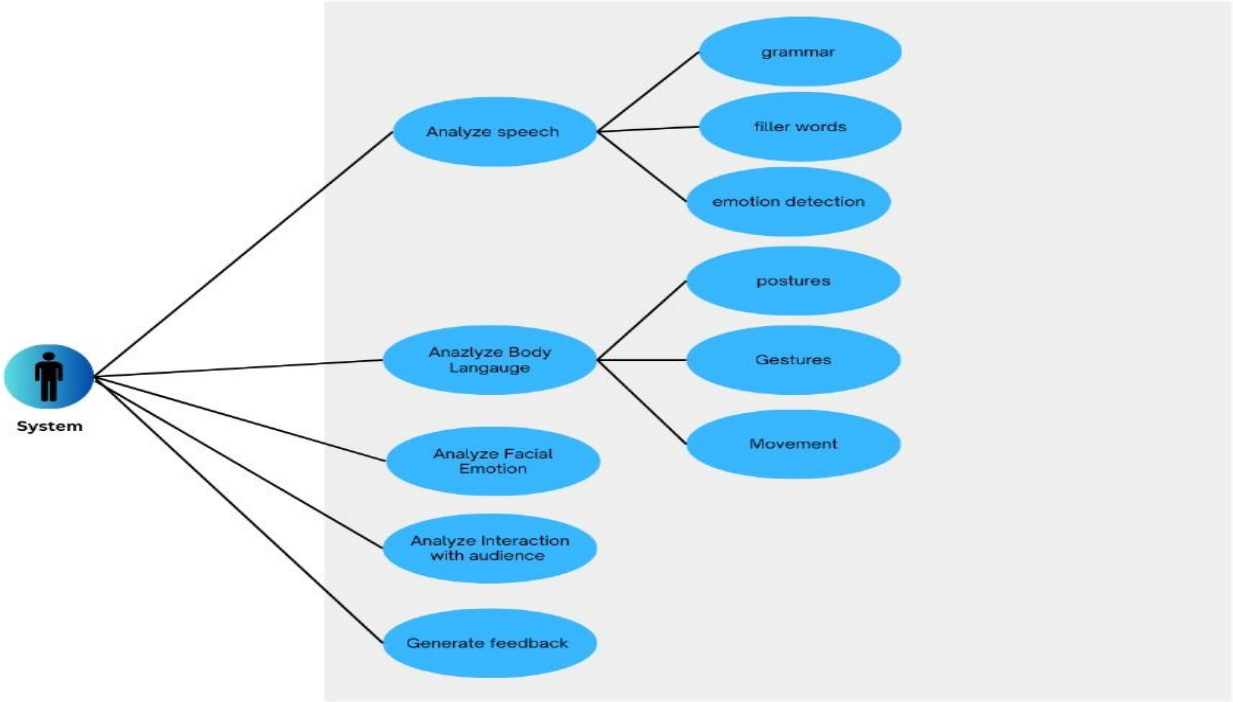
02 REQUIREMENTS

Admin Use Case Diagram



02 REQUIREMENTS

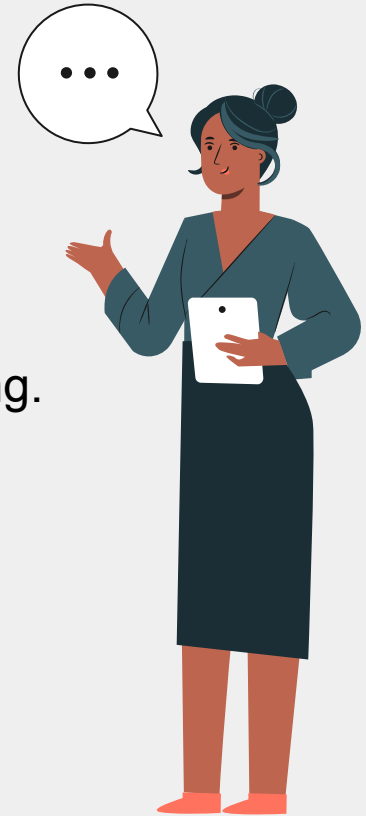
System Use Case Diagram



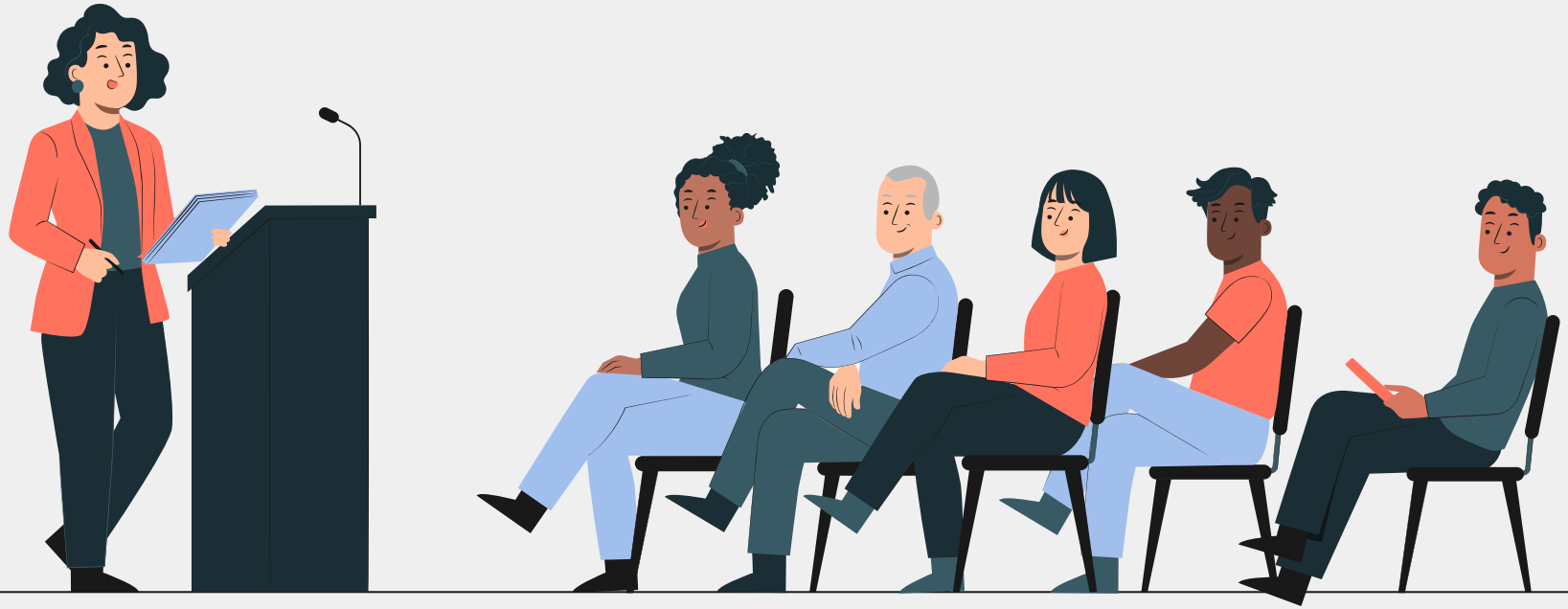
02 REQUIREMENTS

Non-Functional Requirements

Security: Use hashed passwords & role-based access control.
Maintainability: Regular performance monitoring & error logging.
Usability: User-friendly, intuitive interface.
Accessibility: Fully responsive across devices.
Extensibility: Scalable design for future enhancements.



03 METHODOLOGY



03 METHODOLOGY

Algorithms used :

- **Body Language (postures):** PoseNet
  - Captured body keypoints (head, shoulders, hips, etc.)
  - Analyzed alignment to classify postures (e.g., Slouching, Head Up)
  - Mapped postures to engagement levels (e.g., "Leaning Forward" → Interest)

- **Body Language (gestures):** MediaPipe
  - Captured video input
  - Detected hand landmarks (21 key points per hand)
  - Classified gestures based on finger positions (e.g., Open Palm, Thumbs Up)

03 METHODOLOGY

Algorithms used

- **Facial expressions:** Using VGG19 as a base model with added convolutional layers.

- **Tone Detection:** The model is a CNN-LSTM architecture, combining Convolutional Neural Networks (CNNs) for feature extraction and Long Short-Term Memory (LSTM) networks for sequential pattern learning in audio data.

03 METHODOLOGY

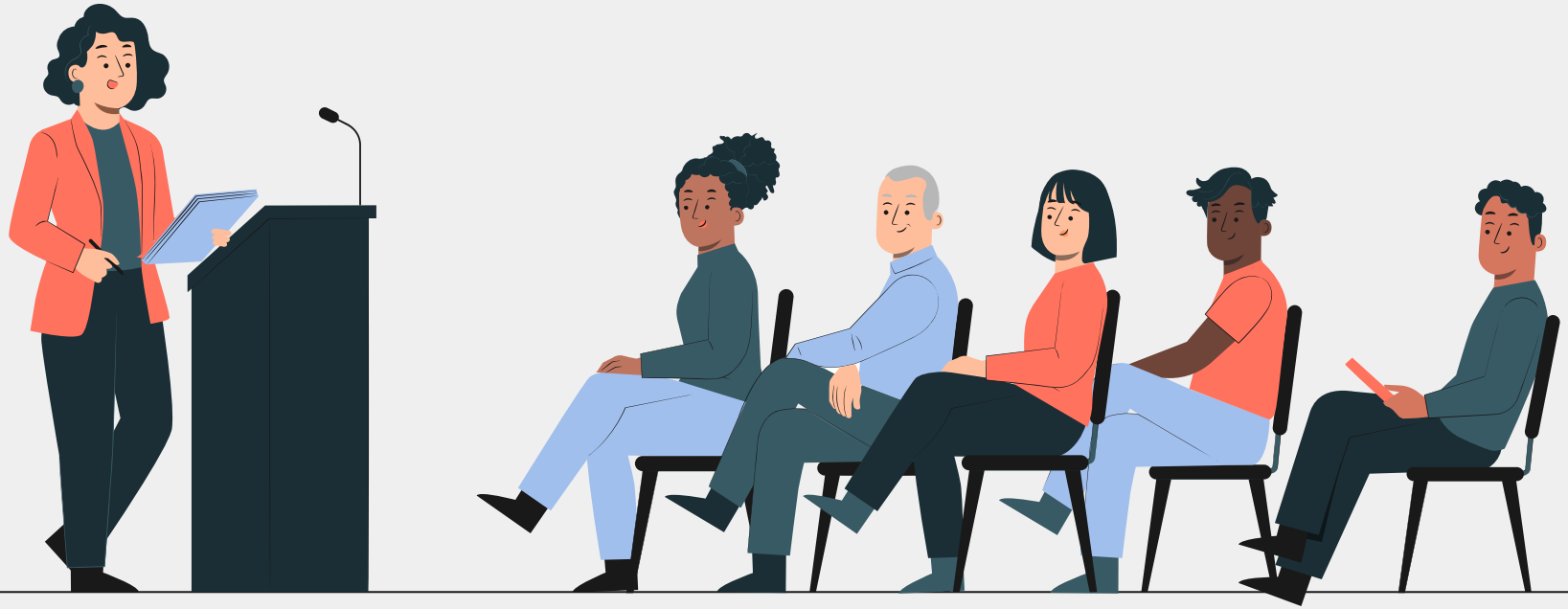
Algorithms used

- **Grammar & Pace:**The Google speech api is used to transcribe from audio to text with timestamps and then the transcription is passed into the t5-base-grammar-correction model.

- **Filler Words:** Facebook wave2vec base model is fine tuned with the podcast fillers dataset which is used to detect filler words.

- **Pronunciation:** A fine-tuned version of wave2vec is used to analyze speech and return an appropriate score.

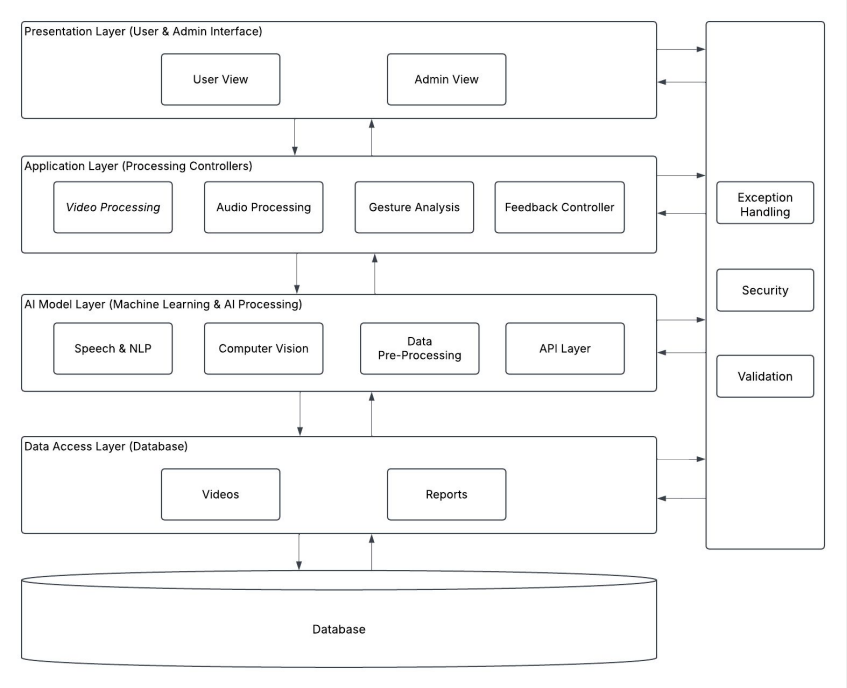
04 DESIGN



04 DESIGN

Architectural Diagram

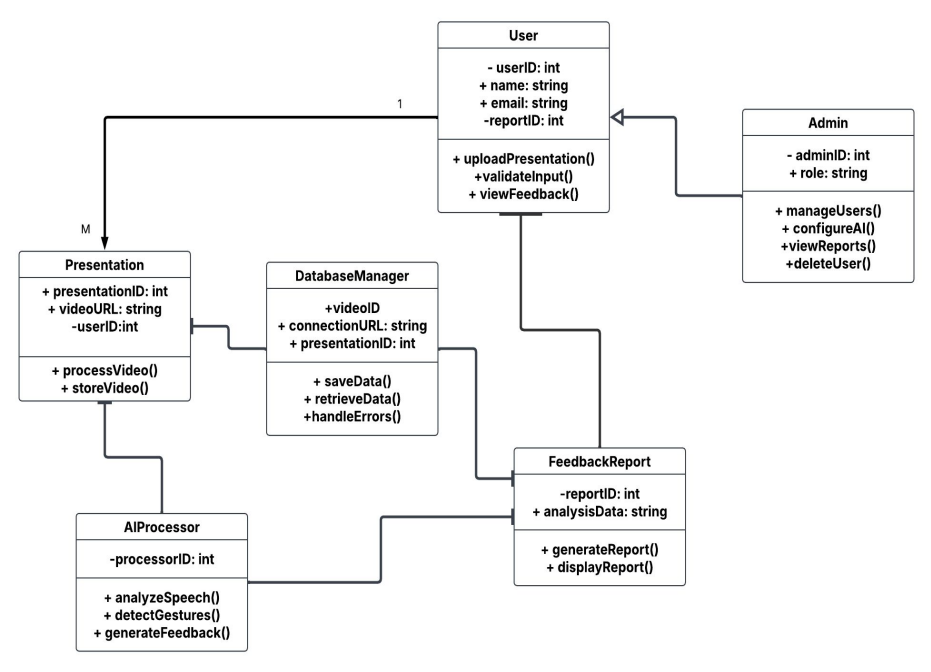
- Layered design ensuring modularity and scalability.
- Layers:
- Presentation Layer: User/Admin interfaces.
- Application Layer: Processes video, audio, gestures, and feedback.
- AI Model Layer: NLP, computer vision, and AI processing.
- Data Layer: Manages video and report storage.
- Security, validation, and exception handling enhance system reliability.



04 DESIGN

Class Diagram

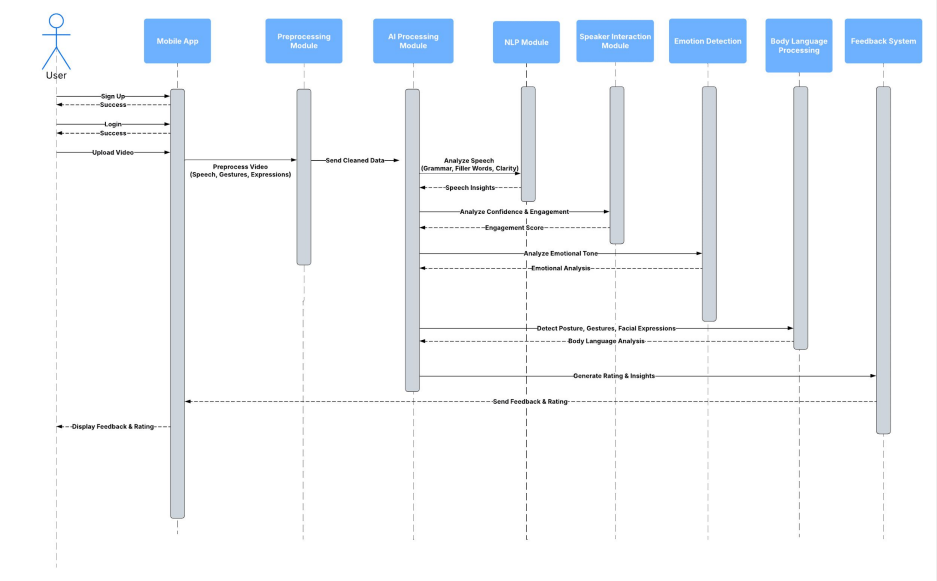
- User: Uploads presentations and views feedback.
- Admin: Manages users and configures AI settings.
- Presentation: Stores video details and processes uploaded files.
- DatabaseManager: Handles data storage and retrieval.
- AIProcessor: Analyzes speech, gestures, and generates feedback.
- FeedbackReport: Generates and displays the final evaluation report.



04 DESIGN

Sequence Diagram

- User signs up, logs in, uploads video.
- Preprocessing cleans data.
- AI analyzes speech, gestures, emotions, and body language.
- Feedback system generates insights.
- User receives evaluation report and ratings.



04 DESIGN

Entity-Relationship Diagram (ERD)

User table:

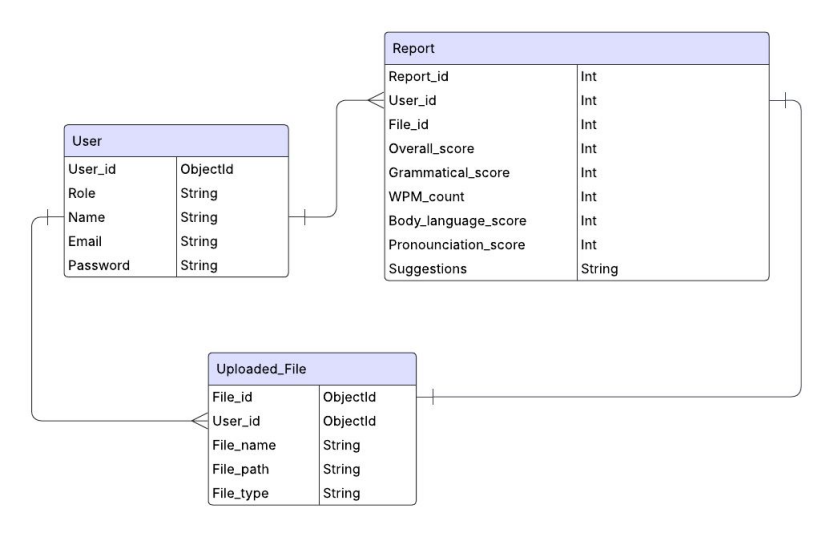
- Stores user information
- Users can upload files and receive reports

Uploade_File table:

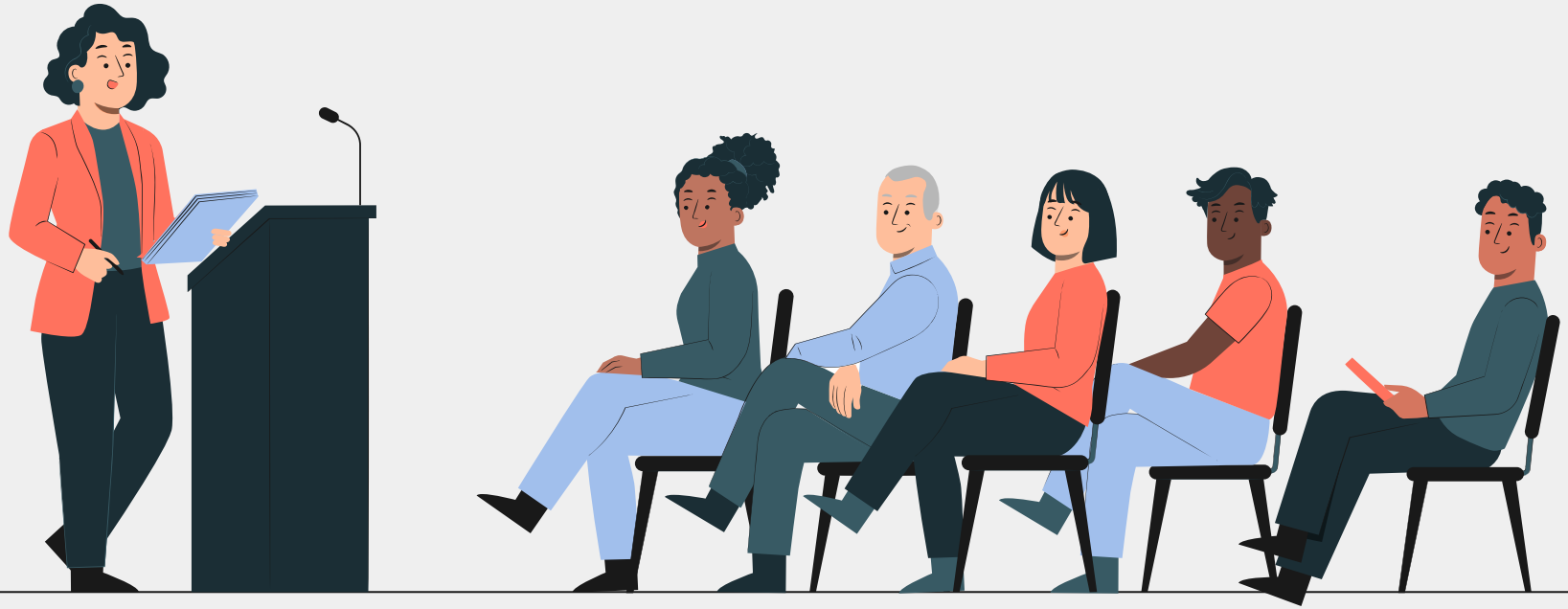
- Stores File information
- Each file is linked to a user and is analyzed for speech report.

Report:

- Stores scores and report details.
- Each report is linked to a user and a specific Uploaded_File



05 USER INTERFACE



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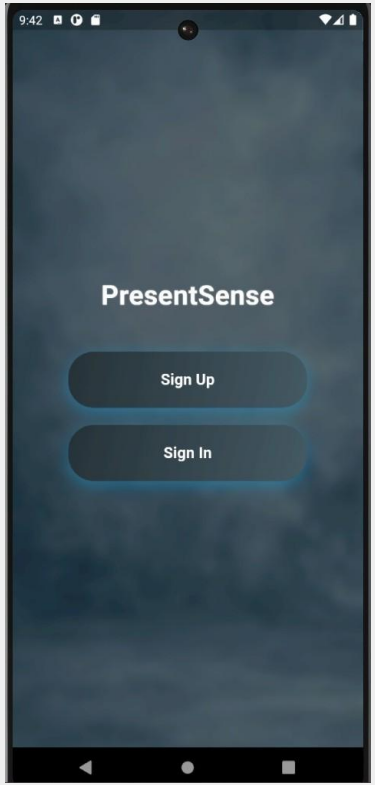
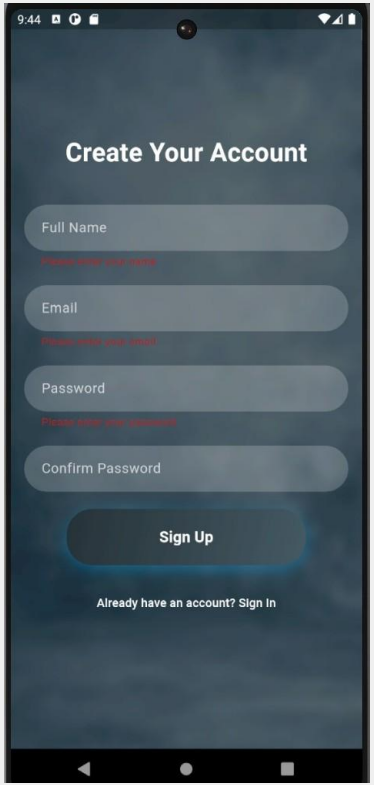
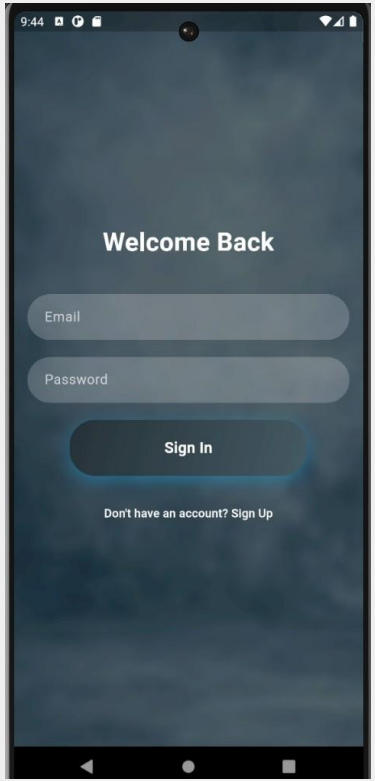
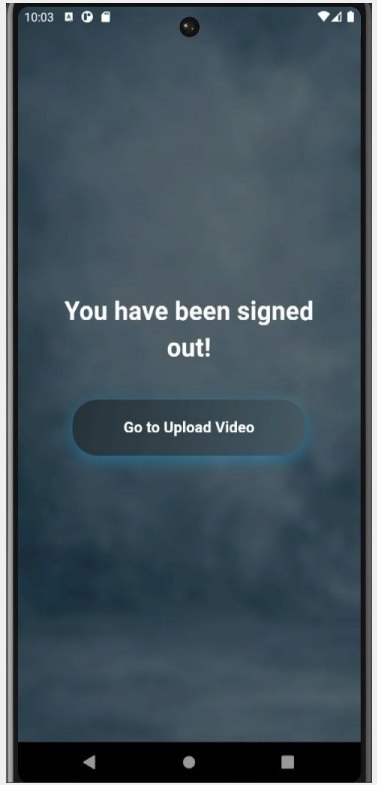
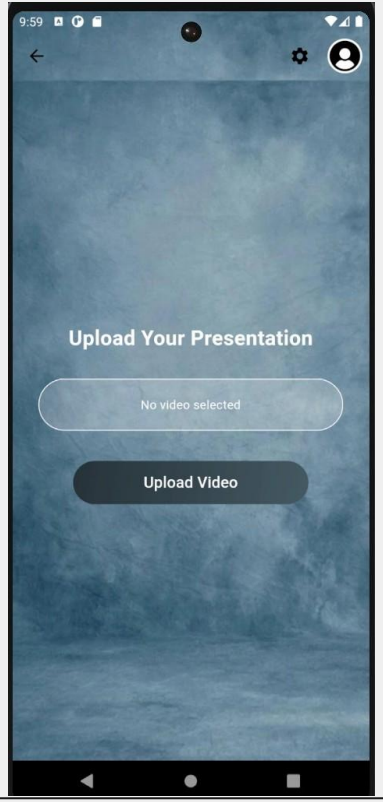


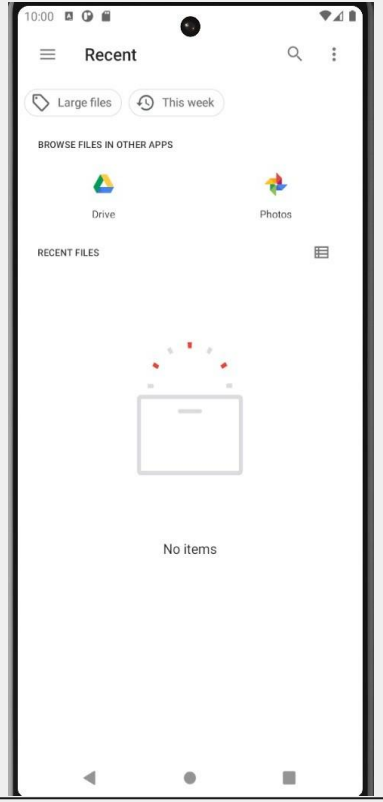


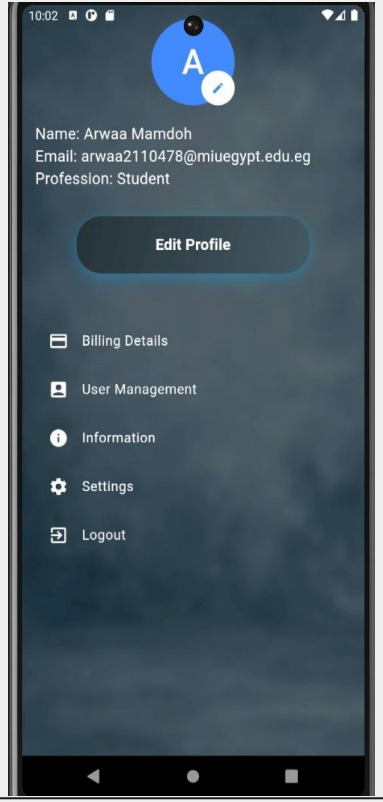


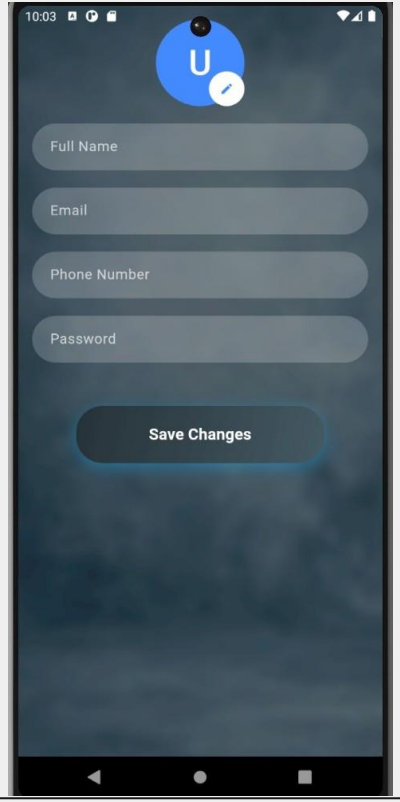


05 USER INTERFACE









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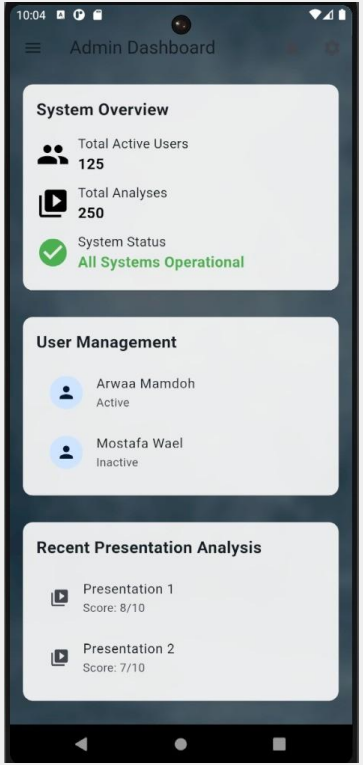
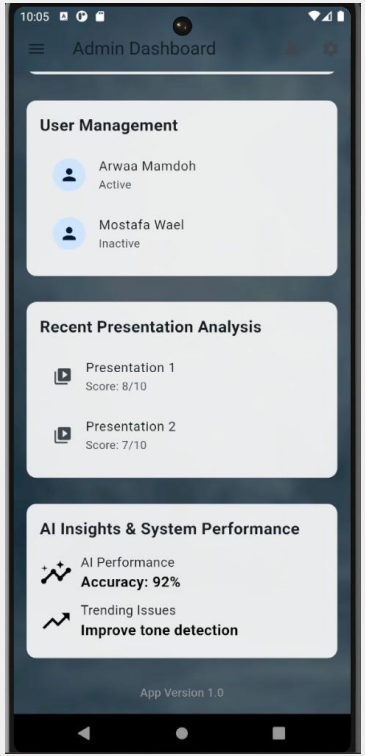
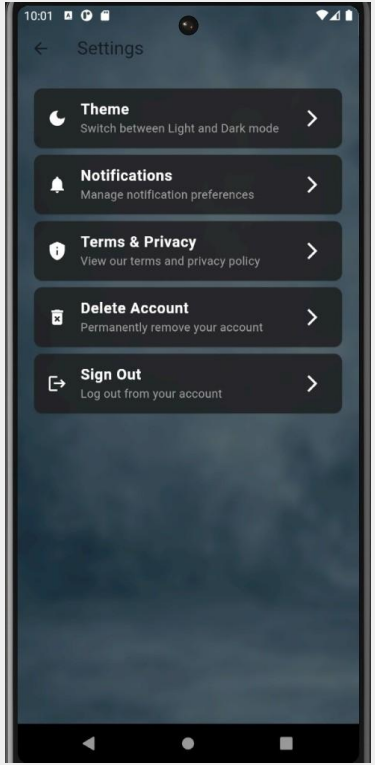
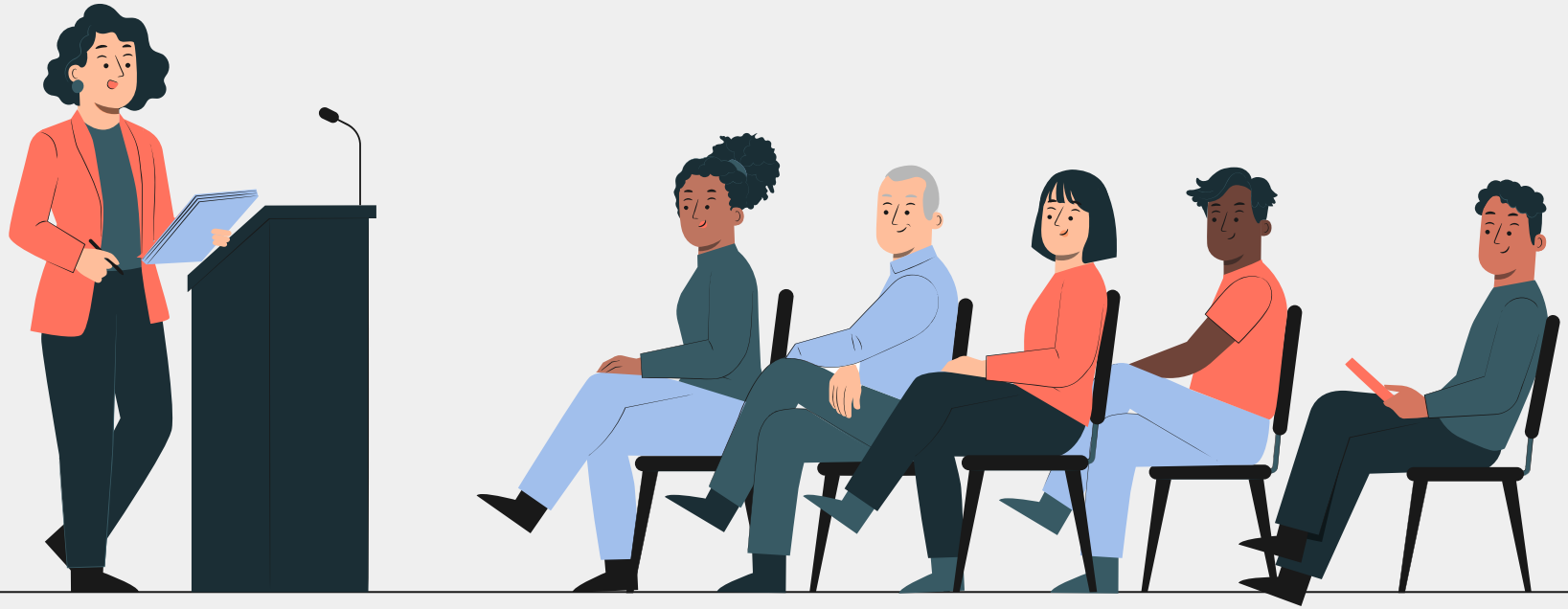






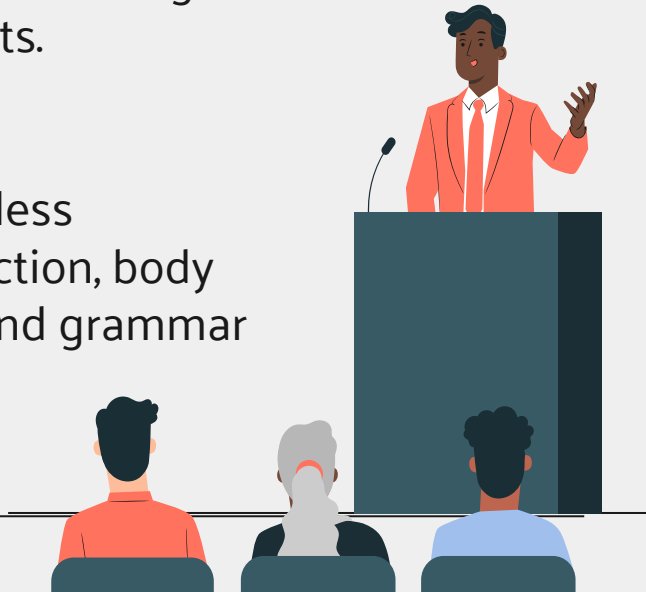
06 WHAT'S NEXT



06 WHAT'S NEXT

- **Improving AI Accuracy** : Refining the AI models for better speech analysis, tone detection, and body language evaluation through additional training and fine-tuning on diverse datasets.

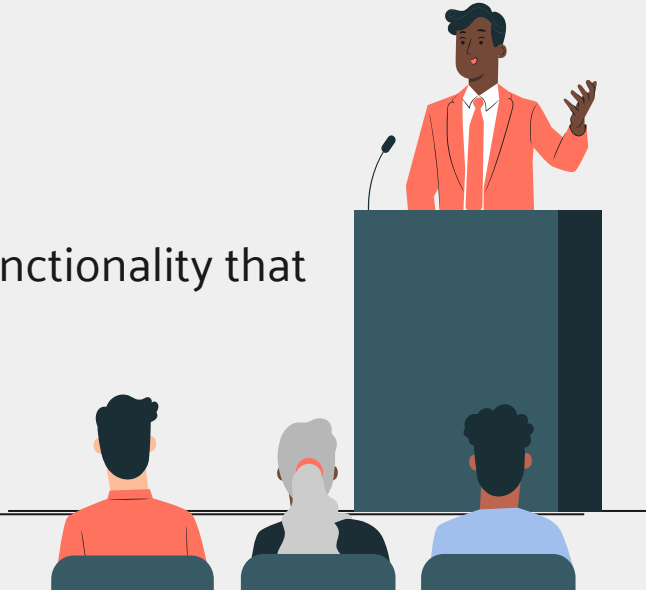
- **Integration of All Models Together:** Ensuring seamless communication between speech analysis, tone detection, body language evaluation, facial expression recognition, and grammar analysis models.



06 WHAT'S NEXT

- **Interaction With Audience :** Upon integrating our models into a unified system, it will have the capability to detect instances of speaker interruption and provide feedback.

- **Application Functionality** : Handling the backend functionality that powers the app.

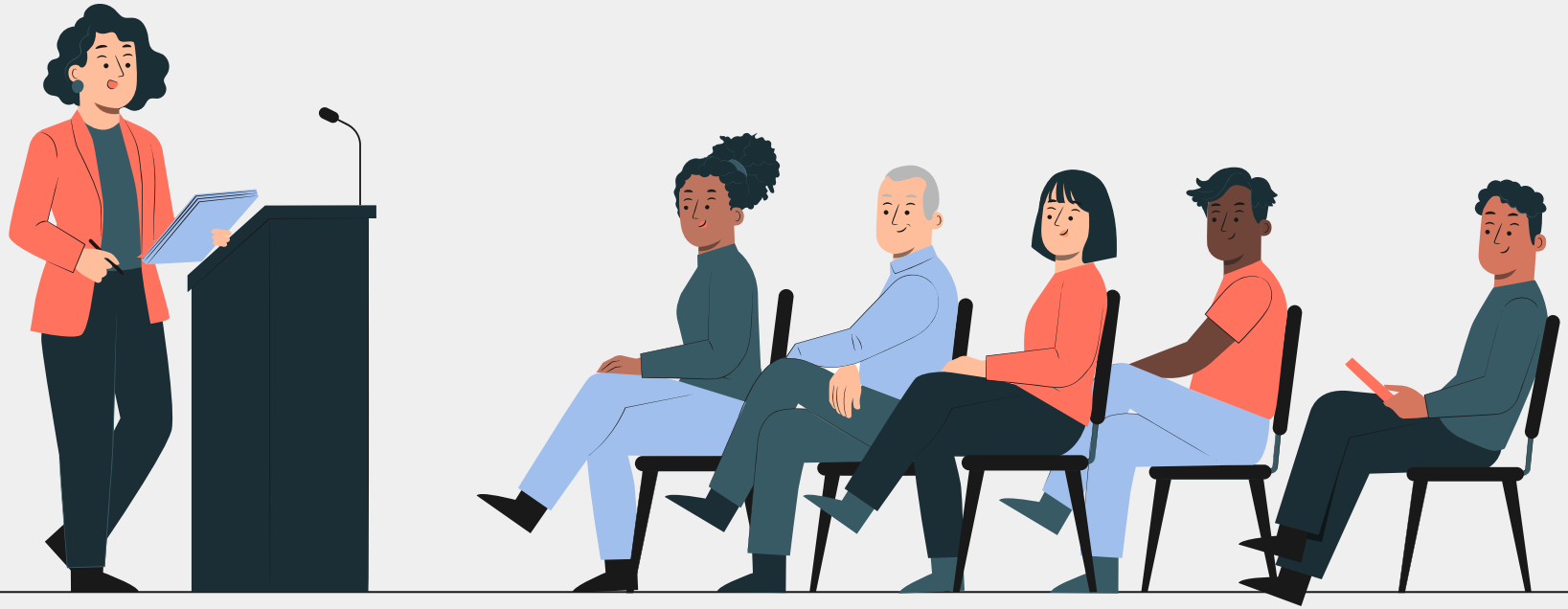


06 WHAT'S NEXT

- **Integration between Ai and App :** integrate the AI models with the app via APIs, enabling seamless communication between the app's frontend and backend . These APIs will handle data processing, analysis, and feedback generation in real-time.

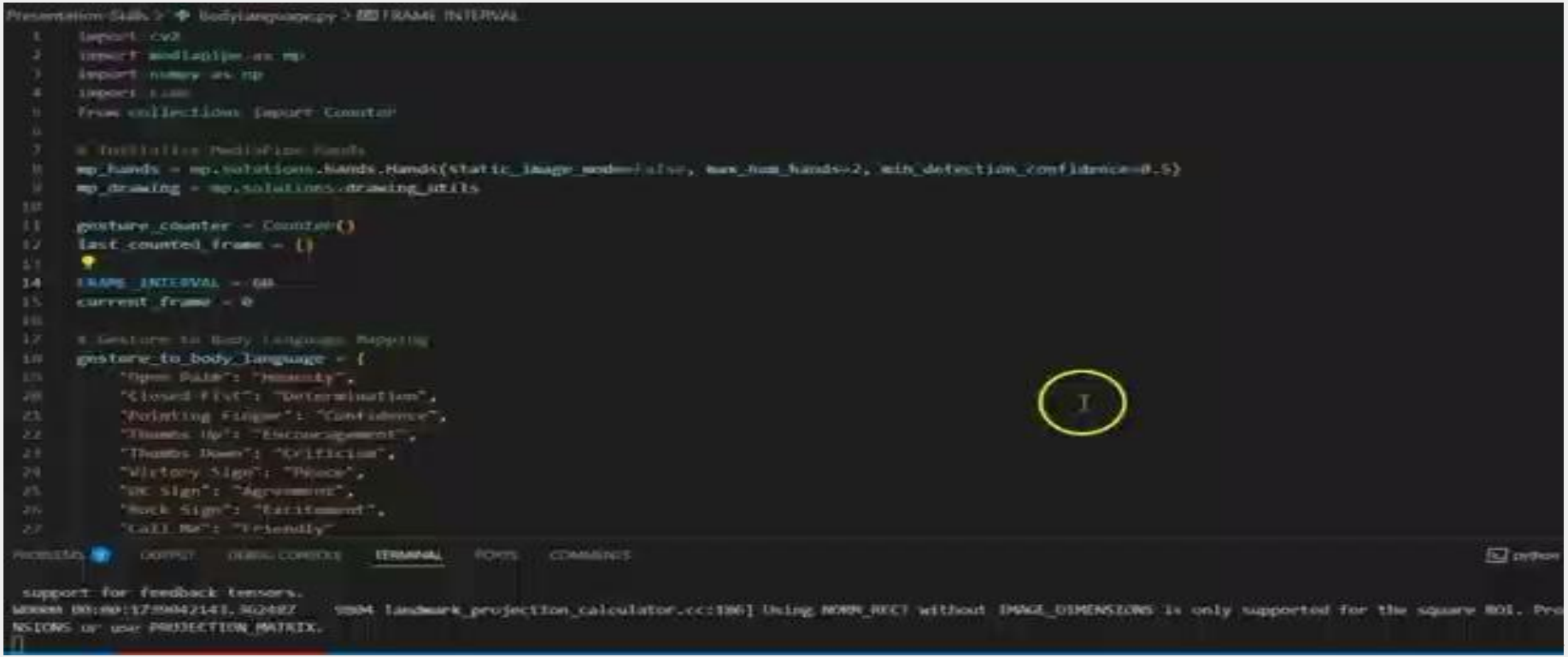
- **Application Deployment :** The app will be deployed on a cloud platform for scalability , It will be available for download on the Play Store and App Store for user accessibility.

07 DEMO



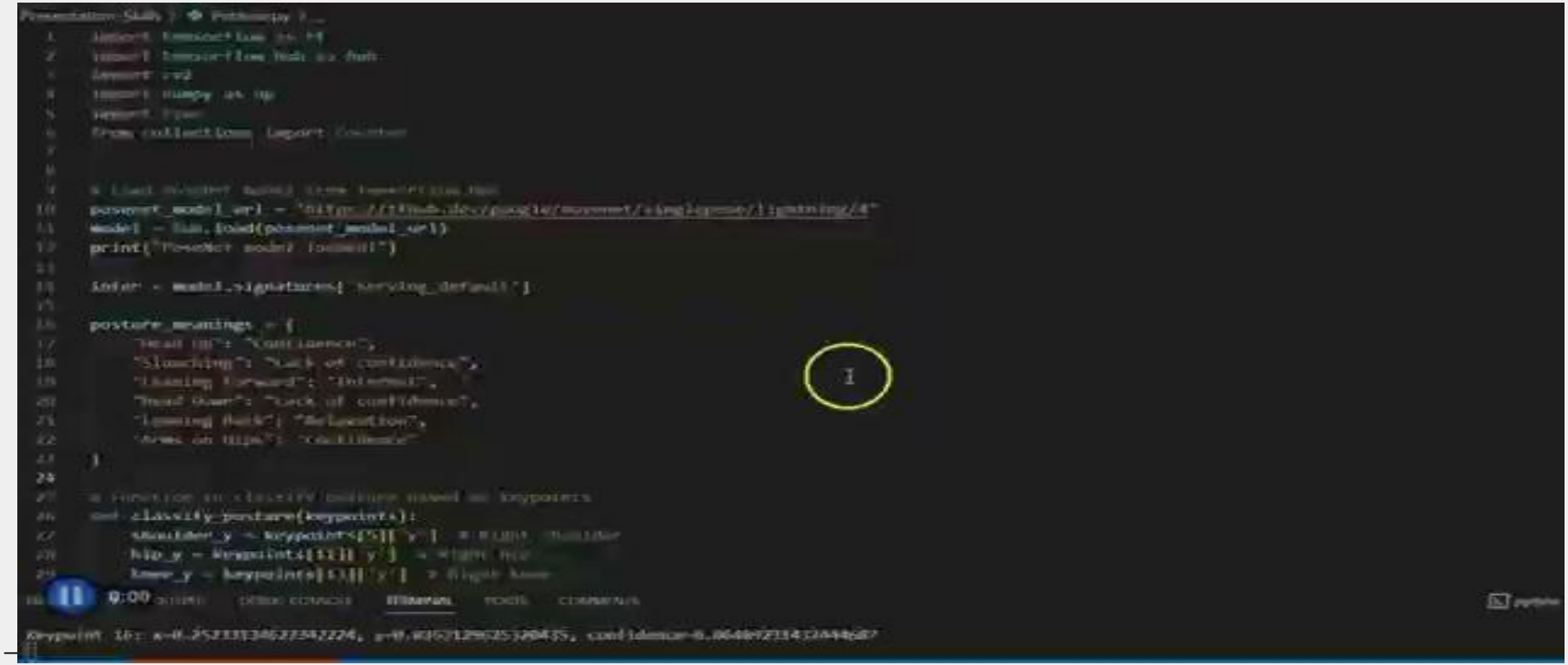
07 DEMO

- **Gesture**



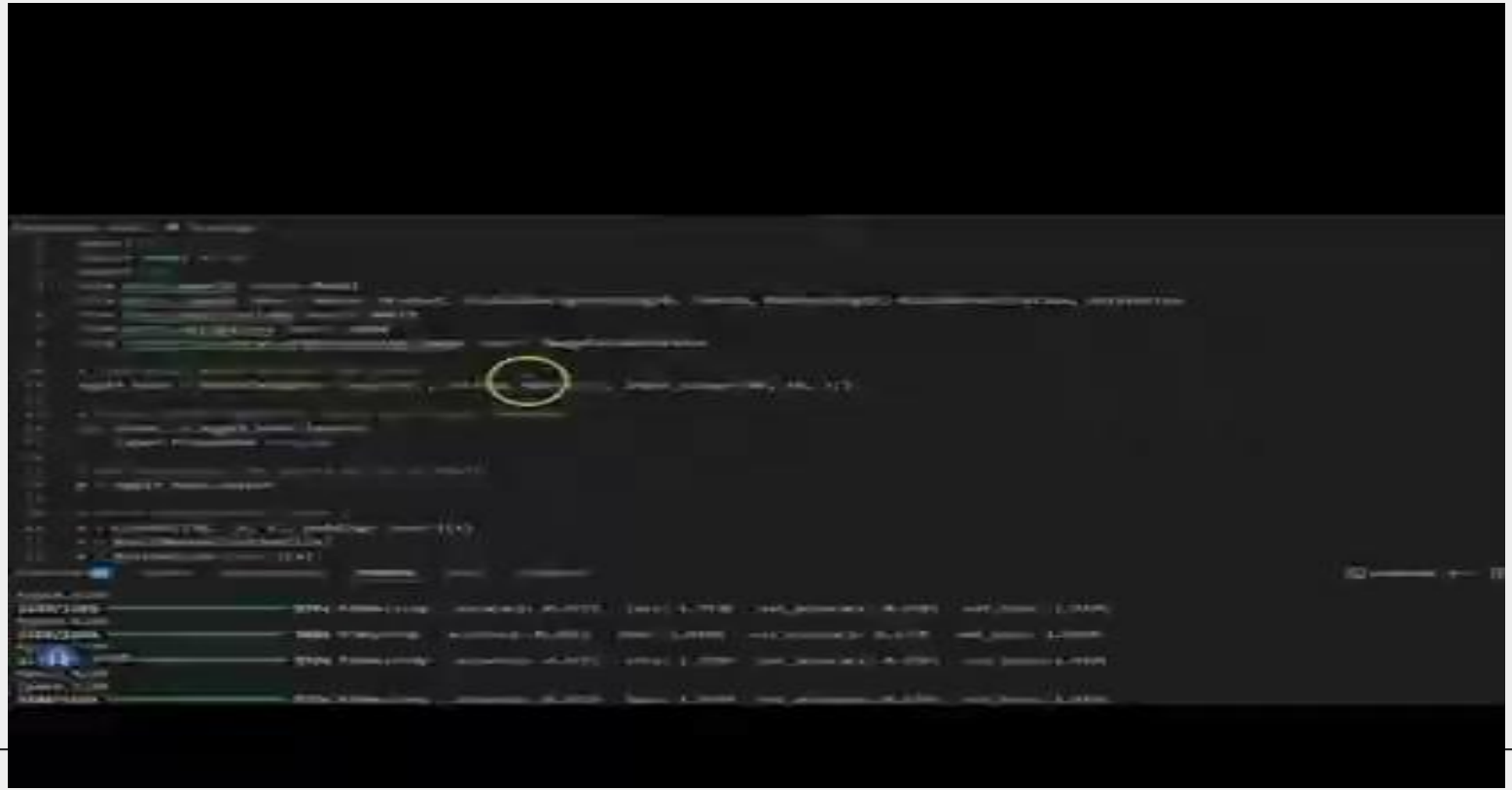
07 DEMO

- **Posture**



07 DEMO

- Facial Emotion



07 DEMO

- **Tone**

File Edit Selection View Go Run Terminal Help Presentation-Skills-1

modified.py 1

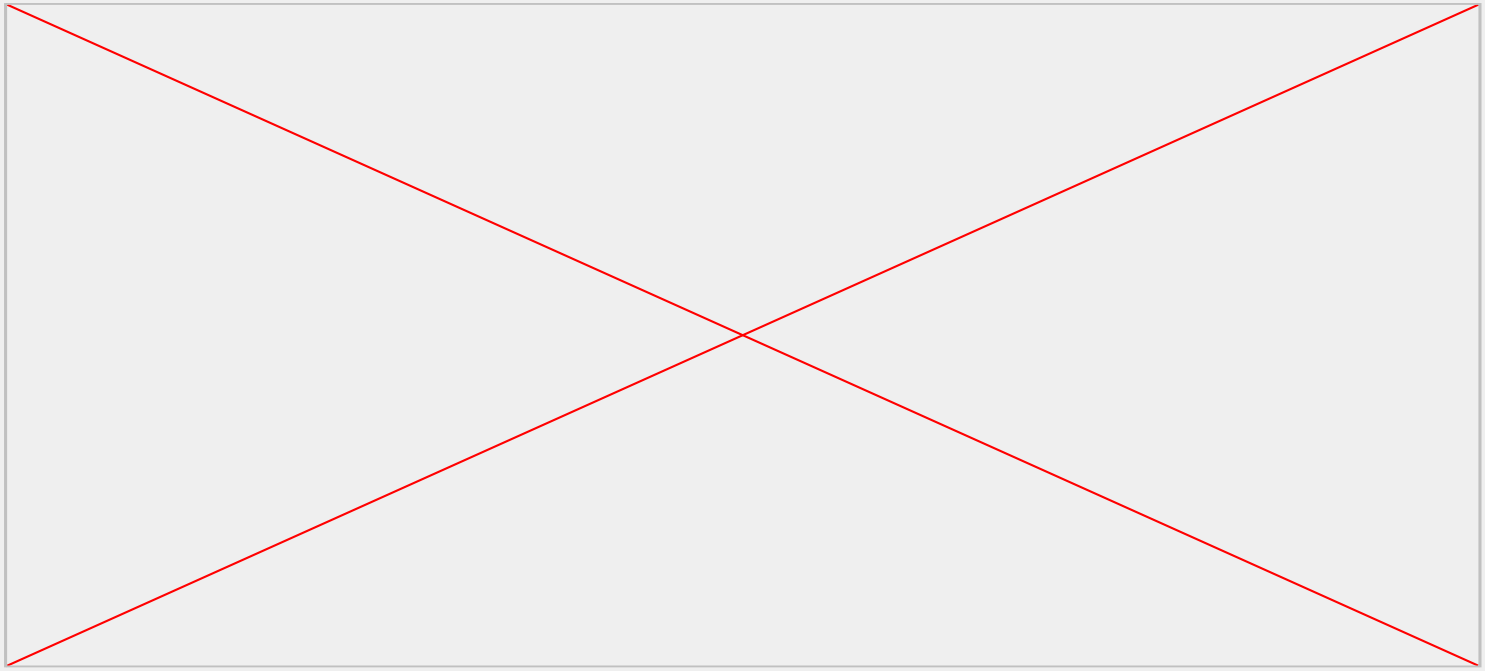
C: > Users > MOUSTAFA > Downloads > modified.py > ...

```python
import numpy as np
import pickle
import librosa
import soundfile as sf
from keras.models import load_model
from sklearn.preprocessing import StandardScaler

#  Load the trained model from .h5 (Corrected path)
model_path = r"C:\Users\MOUSTAFA\Downloads\CNN_Model_Weights (1).h5"
loaded_model = load_model(model_path)

#  Compile the model manually (if needed)
loaded_model.compile(optimizer="adam", loss="categorical_crossentropy", metrics=["accuracy"])
print(" Model loaded and compiled successfully!")

# Load the scaler (Pickle file)
scaler_path = r"D:\Speech Emotion Recognition-20250207T115414Z-001\D__Speech Emotion Recognition-20250207T115414Z-001_Speech Emotion Recognition_Scaler.pickle"
with open(scaler_path, "rb") as file:
    scaler = pickle.load(file)
print("Scaler loaded successfully!")

#  Load the encoder (Pickle file)
encoder_path = r"D:\Speech Emotion Recognition-20250207T115414Z-001\Speech Emotion Recognition\Encoder.pickle"
with open(encoder_path, "rb") as file:
    encoder = pickle.load(file)
print(" Encoder loaded successfully!")

#  Define Model Expected Input Shape
EXPECTED_FEATURES = 2376  # Change this to match your model's expected input size

#  Function to extract features
def compute_zcr(audio_data, frame_length=2048, hop_length=512):
    return np.squeeze(librosa.feature.zero_crossing_rate(audio_data, frame_length=frame_length, hop_length=hop_length))

def compute_rmse(audio_data, frame_length=2048, hop_length=512):
    return np.squeeze(librosa.feature.rms(y=audio_data, frame_length=frame_length, hop_length=hop_length))
```

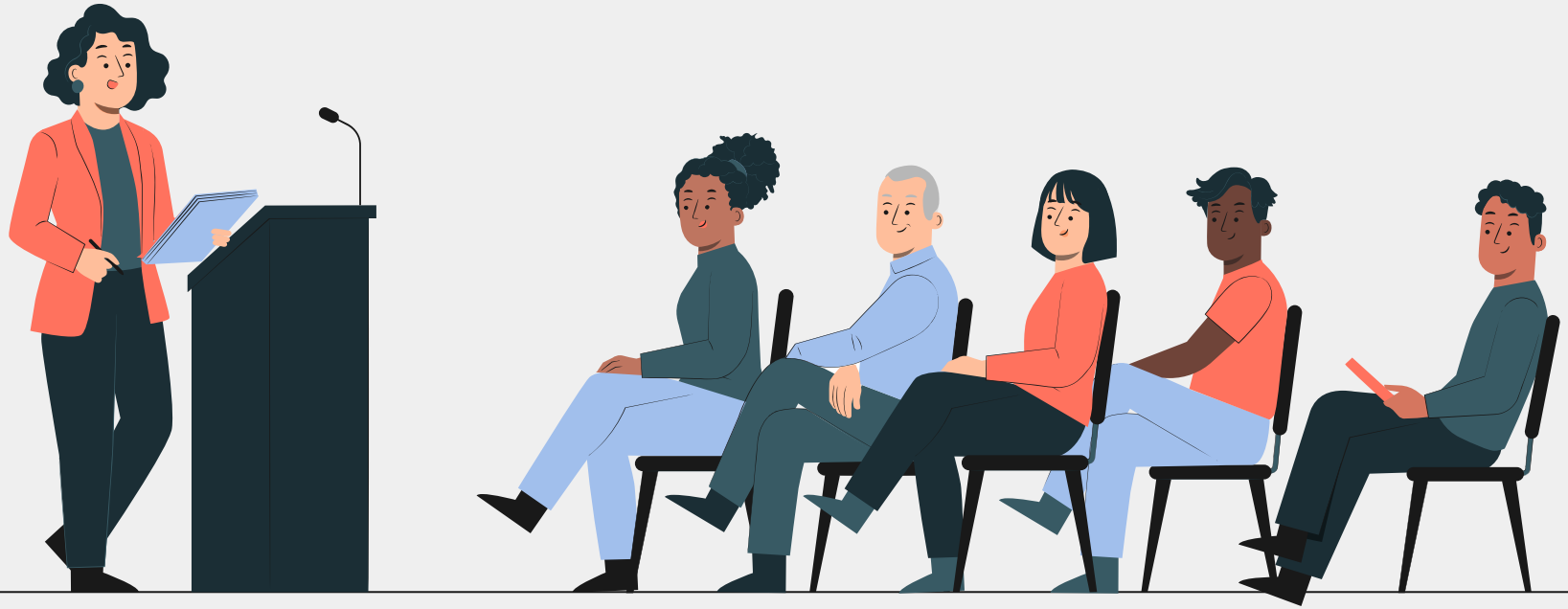
main* 4↓ 0↑ ⊗ 0 ⚠ 24 ⓘ 116 Ln 96, Col 63 Spaces: 4 UTF-8 CRLF {} Python 3.11.0 Windows (windows-x64)

07 DEMO

- **Filler words & Pronunciation**



08 REFERENCES



08 REFERENCES

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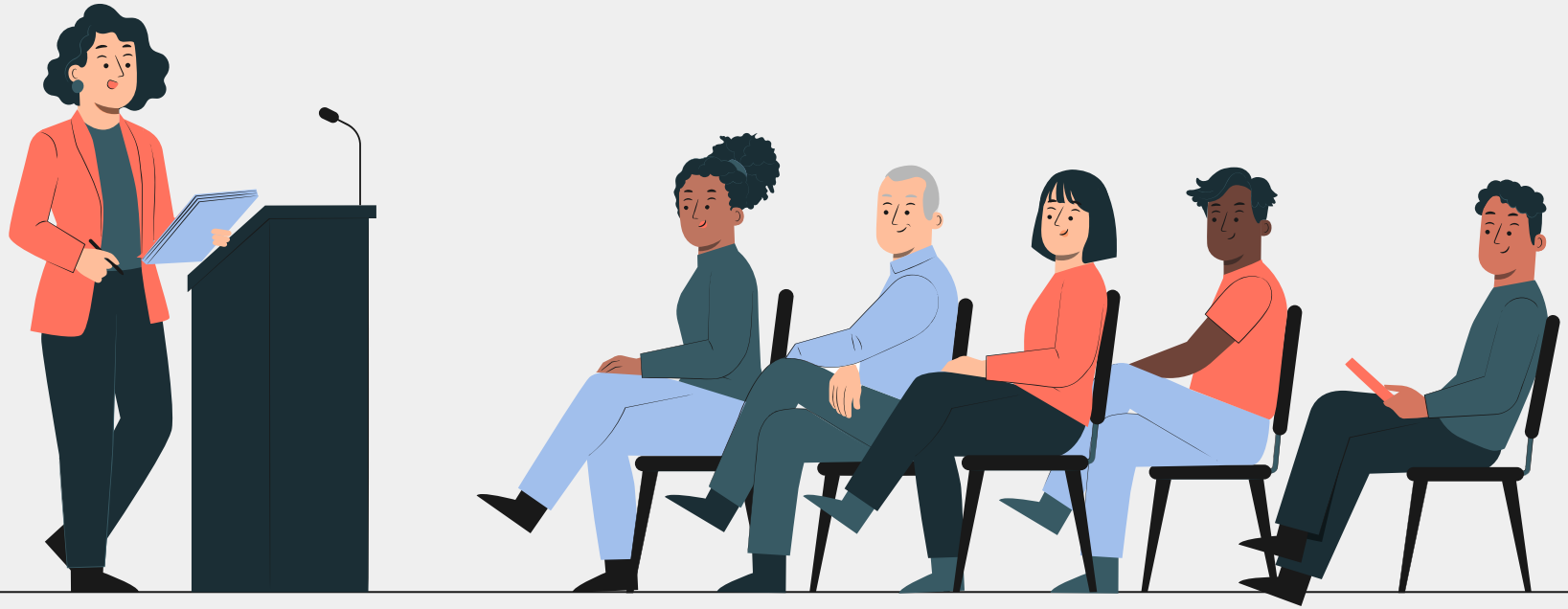
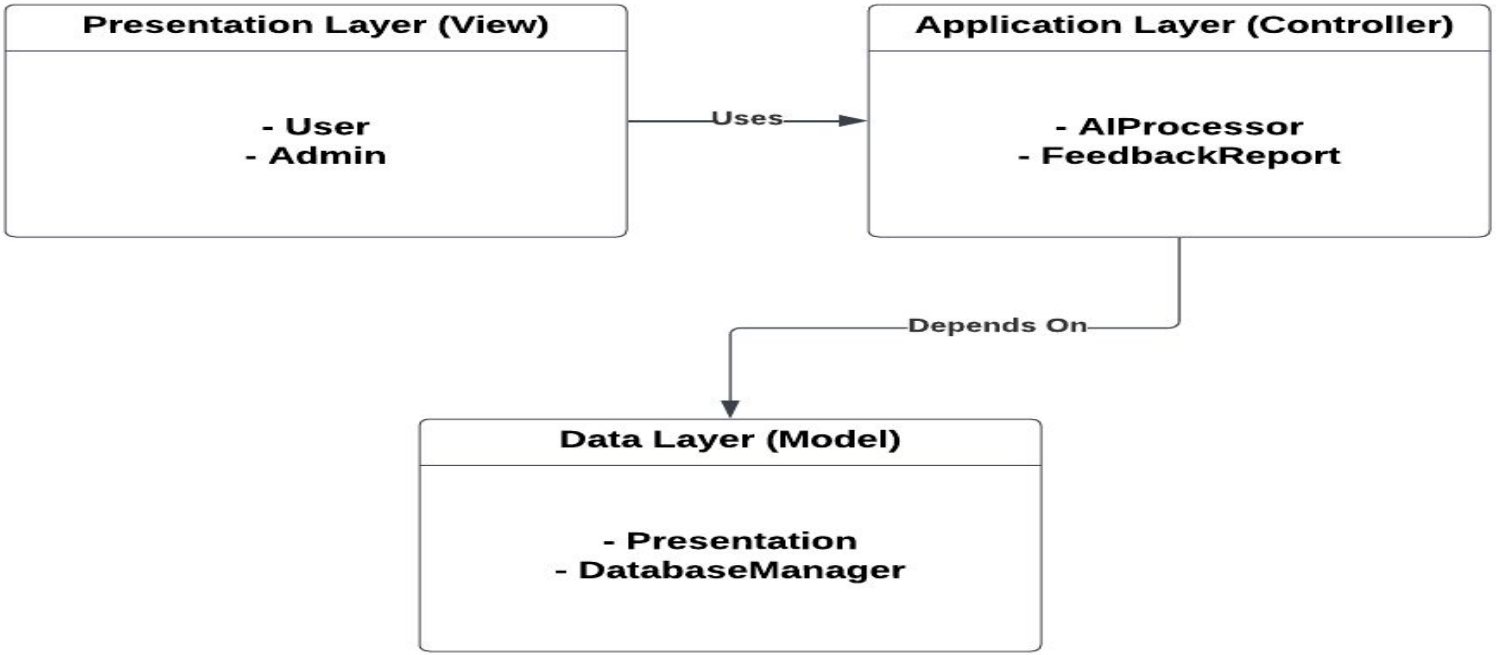
09 APPENDIX



09 APPENDIX

UML Package Diagram



09 APPENDIX

Dataset Description table:

Dataset Name	Number of Samples	Number of Classes	Format
SAVEE	480	7	.wav
TESS	2800	7	.wav
RAVDESS	1440	7	.wav
CREMA-D	7442	6	.wav
FER2013	35,887	7	.png
AffectNet	1M+	8	.jpg

Thanks!

Do you have any questions?